

Official L^AT_EX Beamer Template of the Chair for AI Methodology (**AIM**) RWTH Aachen University Quick Guide

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Introduction

- ▶ Use an introductory slide (or introductory slides) to motivate the topic and to raise interest.
- ▶ You can show an outline of the talk (table of contents) if you want, but we strongly recommend to place it after the introduction and *not* before.

Blocks

Regular block

This is a plain and simple block.

Example block

This is for examples.

Alert block

Use this one to state important information.

Lists

- ▶ Lorem ipsum ...

Lists

- ▶ Lorem ipsum ...
- ▶ dolor sit amet ...

Lists

- ▶ Lorem ipsum ...
- ▶ dolor sit amet ...
 - ▶ Consequetur amibilibisque utero

Lists

- ▶ Lorem ipsum ...
- ▶ dolor sit amet ...
 - ▶ Consequetur amibilisque utero
 - ▶ Anhilore deus et arendum

Lists

- ▶ Lorem ipsum ...
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 - ▶ Anhilore deus et arendum
- ↪ Custom label

Lists

- ▶ Lorem ipsum ...
 - ▶ dolor sit amet ...
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 - ▶ Anhilore deus et arendum
- ~ Custom label
1. Idiquit et collequt deribur

Lists

- ▶ Lorem ipsum ...
- ▶ dolor sit amet ...
 - ▶ Consequetur amibilibusque utero
 - ▶ Anhilore deus et arendum

~ Custom label

1. Idiquit et collequit deribur
2. Canum meum id comedid

Citations with biblatex

Good alternative to natbib

Sample citation: Bossek et al. 2019

Sample citation in parenthesis: (Bossek et al. 2019)

Sample full citation via `\fullcite`

Jakob Bossek et al. (2019). “Runtime Analysis of Randomized Search Heuristics for Dynamic Graph Coloring”. In: *Proceedings of the 21th Genetic and Evolutionary Computation Conference*. GECCO '19. Prague, Czech Republic: ACM.

Images

Use macro `\ig{width}{path-to-file}` for a single **centered** image:

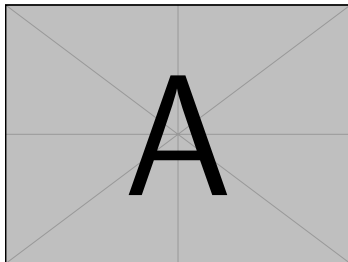
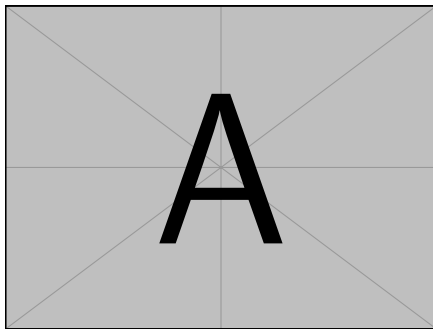


Figure environment

Feel free to use the figure environment. Note that in the captions (also for tables) our template deliberately omits the label 'Figure:' before the caption:



test

Tables

We recommend to take a look at the presentation [Small Guide to Making Nice Tables](#) by Markus Püschel.

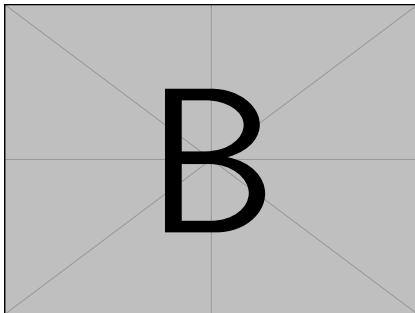
TSP Set	Mutation	RTS [*]		FR [†]		PAR10	
		EAX	LKH	EAX	LKH	EAX	LKH
RUE	-	1.26	0.74	0.00	0.00	1.26	0.74
Easy for	simple	1.34	912.78	0.00	0.20	1.34	7 608.11
EAX	sophistic.	0.97	830.80	0.00	0.22	0.97	8 230.61
Easy for	simple	117.97	0.74	0.00	0.00	117.97	0.74
LKH	sophistic.	67.90	0.88	0.00	0.00	67.90	0.88

^{*} RTS: Running time of successful runs, [†] FR: Failure ratio

Columns

The `\twocol{...}{...}`-macro aligns stuff on top and uses two equally sized columns:

- ▶ Lorem ipsum
- ▶ dolor sit amet
- ▶ consequetur deribilis auret
 - ▶ Alamat deceductovo ameritol
 - ▶ Consequencias pavit



Math

... looks awesome in L^AT_EX

Sample formula

Math looks so awesome in L^AT_EX!

$$\hat{\theta}_{\text{ML}} = T(X_1, \dots, X_n) = \frac{n}{n-1} \sum_{\substack{i=1 \\ i \neq k}}^m (X_i^2 - \exp(X_i - X_k))^{k/2}$$

Theorem (Bossek et al. 2019)

For every tree with n nodes and maximum degree Δ the expected time until RLS and $(1+1)$ EA find an optimal Δ -edge-coloring is $O(\Delta \ell^2 m \log m)$ where ℓ is the length of the longest path in the tree.

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Pseudo-code

... with algorithm2e

Algorithm 1: Sample algorithm (taken from Overleaf docs)

Input: $n \geq 0$

Output: $y = x^n$

1 $y \leftarrow 1, X \leftarrow x, N \leftarrow n;$

2 **while** $N \neq 0$ **do**

3 **if** N *is even* **then**

4 $X \leftarrow X \times X;$

5 $N \leftarrow \frac{N}{2};$

// This is a comment!

6 **else**

7 **if** N *is odd* **then**

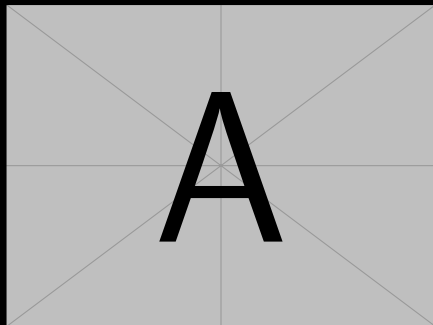
8 $y \leftarrow y \times X;$

9 $N \leftarrow N - 1;$

Useful macros

- ▶ Use `\refer{...}` to refer a paper quickly (without the need for a bibfile entry): Bossek et al., 2019
- ▶ Use `\hide{...}` to temporarily hide block of code.
- ▶ Use `\plainframe{...}` for a visually reduced frame with horizontally and vertically centered message/image (see next slides for examples).
- ▶ Use `\plainframe[mycolor]{...}` to change the default background.

Plain slide (focus on certain element)



Commenting macros

It is often useful to comment on different things while writing a report, adding Todos or highlight changed or added parts. To this end the file `includes/commenting.tex` defines some useful macros.

- ▶ Use `\todo{...}` to add a ToDo:

TODO ▶ *Do this, do that*◀

- ▶ Use `\changed{...}` to indicate changes:

CHANGED ▶ *This text was changed.*◀

- ▶ Use `\added{...}` to highlight additions:

ADDED ▶ *This text was added.*◀

- ▶ Use author-specific macros, e.g., `\jane{...}` or `\john{...}` for our two sample authors Jane and Joe, to add comments. Feel free to edit `includes/commenting.tex` to add/adapt the author-specific macros.

Jane ▶ *Comment by Jane.*◀

John ▶ *Comment by Joe.*◀

Take-home message

- ▶ Briefly summarise the main finding. What are the most important aspects the audience should keep in mind?
- ▶ Your presentation should never end with a slide showing the references or a plain slide with *Questions?* I.e., our recommendation is to end the presentation with the take-home messages.

References I

Bossek, Jakob et al. (2019). “Runtime Analysis of Randomized Search Heuristics for Dynamic Graph Coloring”. In: *Proceedings of the 21th Genetic and Evolutionary Computation Conference*. GECCO '19. Prague, Czech Republic: ACM.

Backup slides

Sequence of additional slides. Useful to keep more information in the background which can be revealed during discussions.